Name: ____________________ Worksheet

Using dynamic software to explore linear relationships

Task A: Changing the gradient

Dynamic software should be used to complete this task.

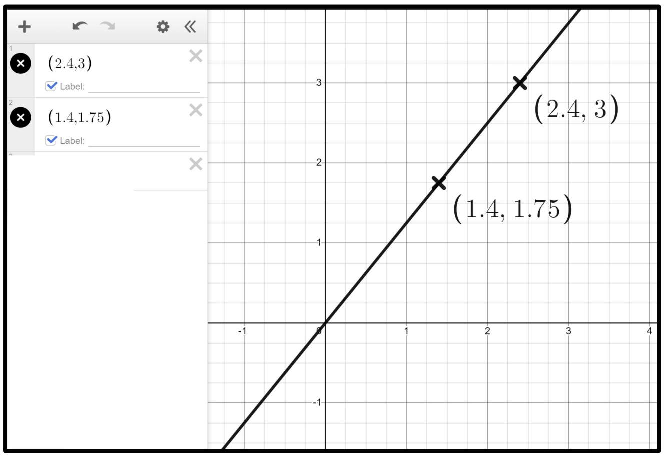
1) You are the teacher.

Plot the line $y = 3x$, add a table, turn the coordinates into crosses, and explain to the Oak students the link between the algebra, the table and the graph.

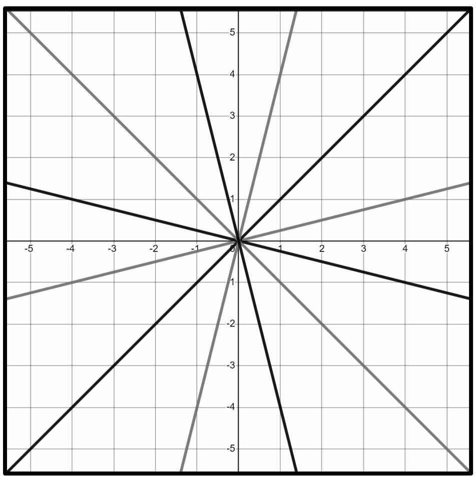
2) Plot the lines $y = -4x$ and $y = \frac{1}{4}x$

Compare the two lines.

3) Plot the coordinates (2.4, 3) and (1.4, 1.75) and draw the straight line that goes through both and the origin. What is this line?



4) Draw the lines that recreate this pattern.



Task B: Changing the y-intercept

Dynamic software should be used to complete this task.
1) Plot the lines and write a sentence or two comparing them.

$y = 2x$

$y = 2x + 3$

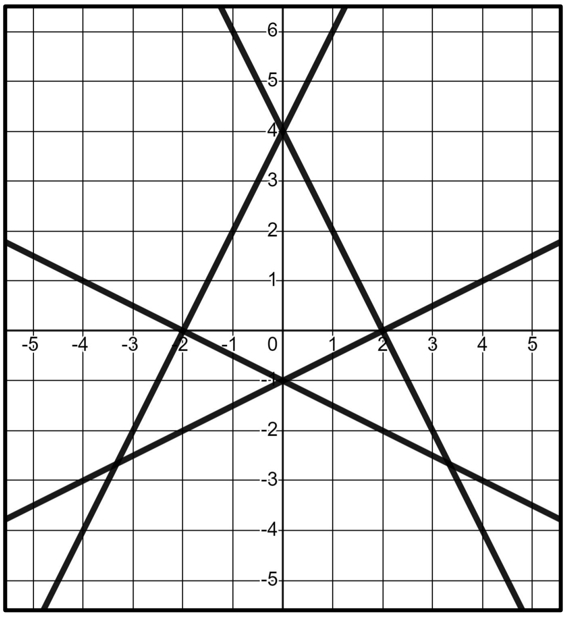
$y = 2x - 4$

2) Plot the lines and complete the table.

Line	Gradient	y-intercept
$y = 3x + 7$		(0, 7)
$y = 3x - 2$		
$y = 0.2x + 2$		
	7	(0, −3)
	−3	(0, 7)

3) Draw the lines that recreate these patterns.

a)



b)